

SAFER: SYSTEM FOR DISASTER ASSESSMENT AND FORECASTING OF EMERGENCY RISKS USING GEO-SPATIAL DATA

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Abstract—Disaster management in Northeast India faces challenges like recurrent natural disasters, inadequate infrastructure, and limited real-time data. So this requires effective disaster management, assessment and accurate forecasting to ensure safety and minimize risks. This paper introduces the “System for Disaster Assessment and Forecasting of Emergency Risks using Geo-Spatial Data” (SAFER), a web and app-based platform that uses advanced geospatial technologies for accurate disaster assessment of floods and landslides using satellites like Sentinel-1 and Sentinel-2. For disaster prediction it integrates a Long Short-Term Memory (LSTM) based model. The system includes a voice-controlled geospatial maps, real-time alerts, and an intelligent ambulance booking system with multilingual support. The SAFER also has a voice-controlled interface which allows users to speak and receive spoken responses, improving accessibility and supporting better emergency response through real-time alerts, reports, and analytics.

Keywords—Disaster Management, Geographic Information System, Long Short-Term Memory, Raster data, Vector data, Floods, Landslides

I. INTRODUCTION

The Northeastern region of India is highly vulnerable to floods, landslides, earthquakes, and cyclones. The high terrain, convoluted river systems, dense forests, with heavy rainfall intensify the impact of these disasters. The monsoon season is critical often causing severe floods and landslides. It is also the time when states like Assam and Arunachal Pradesh routinely face the displacement of thousands of people alongside the destruction of homes, agricultural lands, and infrastructure. Landslides further cut off access to certain communities by blocking roads. Climate change aggravates these problems by causing weather patterns to behave erratically, resulting in more frequent and intense disasters, further amplifying the urgency for the effective management of disasters.

To overcome these obstacles, the proposed System for Disaster Assessment and Forecasting of Emergency Risks using Geo-Spatial data (SAFER), which enhances disaster forecasting and response capacities. Real-time data that is integrated with advanced geospatial data and early warning systems provide the SAFER with precise assessment and forecasting. User-interface features include voice command-enabled map interface, real-time alerts, intelligent ambulance booking system; thus improving preparedness by pre-incident planning, efficient resource dispatching, and optimum emergency response.

While there exist many implementations in the real world, most of them lack in full mobile app solution. Such an example is the North Eastern Regional Node for Disaster Risk Reduction (NERDRR), which focuses more on offering live disaster updates and alerts but fails in some critical aspects in efficient disaster management and response. While the application provides awareness, it lacks features like ambulance booking and emergency first aid.

The paper further discusses an overview of existing system. Thereafter, the proposed system architecture of SAFER will be described. The results and analysis section will critically evaluate the performance of SAFER. Finally, the conclusion and future scope will be discussed and highlighted to ensure the impact and potential for the system to create improvements in future disaster management practices.

II. RELATED WORKS

In this section, we review existing research related to the analysis and forecasting of floods and landslides, geospatial dashboards, ambulance booking system, and voice-controlled geospatial maps drawing on insights from referenced works to contextualize our contributions.

A. Flood

Satellite data and machine learning have advanced flood monitoring. According to Du et al. (2021), flood mapping of Cyclone Idai was carried out from SMAP and Landsat data [5]. Combined data and machine learning models were developed by Puttinaovarat and Horkaew (2020) [12] with an accuracy of 97.93%. Kuldeep, P. Garg and R. Garg. (2016) mapped flood-affected areas by using Indian satellites [6]. Some challenges include data dependency and localization.

B. Landslide

Accuracy is enhanced in landslide detection by using remote sensing and machine learning. According to Chamoli et al. (2024) 94.75% accuracy was achieved through Sentinel-1 SAR data [2] but it failed with low Kappa scores. The “Landslide detection using deep learning and object based image analysis” (2022) reported improvements in F1-scores with ResU-Net [7], while other employed XGBoost to improve interpretability [3]. False positives and generalization are the problems.

C. Statistical analysis

Praharaj and Wentz introduced a toolkit in “Building Community Resilience Through Geospatial Dashboards” [11], which integrates data from multiple agencies into real-time decision-making. Comparing the above dashboards with SAFER, it is realized that the problems of limited data and complex interface are overcome here by including local data, using an easy interface.

D. Intelligent ambulance booking

The innovations in ambulance service amenities, aided by GPS and deep learning, make the services very efficient. “The ambulance booking and tracking system” (2024) designed the web application to book ambulances with GPS guidance, but had issues with respect to the network [1]. Also recent systems brought coordination but faced problems with regard to scalability [8]. While another developed a traffic route optimization system but faced payment and connectivity challenges [13].

E. Voice controlled geospatial map

Communication through voice interface based on voice recognition and geospatial devices. Mahmoudi, Camboim, and Maria in 2023 developed a geospatial voice assistant, whose bases are too complex for a single user and dependent on the Internet [10]. The proposed voice-guided systems can directly assist the visually impaired, but they rely on pre-stored responses created in 2020, which is erroneous and distasteful [9].

III. PROPOSED ARCHITECTURE

The SAFER architecture in Fig. 1 illustrates a system for disaster management with geospatial data, predictive analytics, and real-time data processing. Data collection, integration, processing, and data analysis thus support multiple levels of the architecture, projecting accurate disaster assessment

and forecasting. Moreover, satellite imagery is combined with meteorological data and geographical data to get an overall picture of the situation.

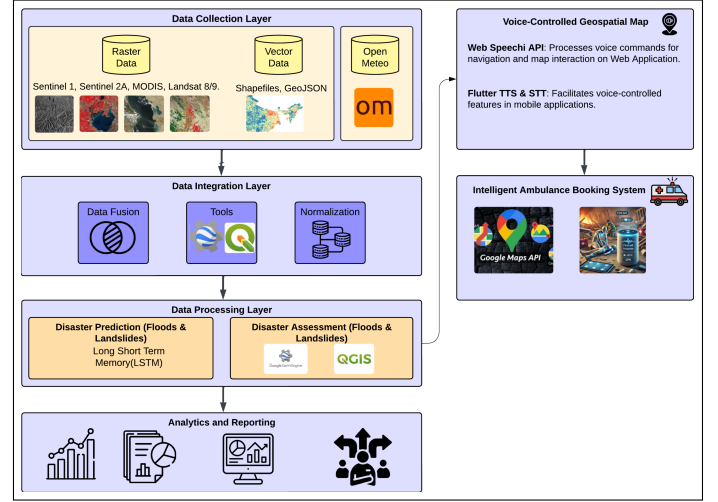


Fig. 1. Proposed system architecture for SAFER

3.1 Data Collection Layer

This layer is the backbone of the system, combining several geospatial and weather datasets, including raster or vector data:

- **Raster Data:** Raster data includes Global Surface Water Dataset, SRTM DEM, Sentinel 2 Surface-Reflectance and Sentinel 1.
- **Vector Data:** Vector data captures geographic features like administrative boundaries. The IND adm2 dataset maps district boundaries of India, thus helping identify flood-prone regions.
- **Open Meteo Data:** Meteorological data was collected from the Open-Meteo API. The data has historical weather and flooding location from these lands that will be studied for analyzing the climate of the flood-prone areas of India.

3.2 Data Integration Layer

Data integration combines vector datasets such as that of Dhuburi to perform filtering on raster data and clip it to focus the analysis on specified regions. By normalization, Lee filtering reduces speckle noise and maintains essential features of images by modifying the pixel values based on local attributive statistics. A focal_median function applies a 30 meter radius circular median filter to SAR images, which makes them quite appropriate for flood analysis.

3.3 Data Processing Layer

The integrated data undergoes further advanced process in this layer:

- **Disaster Assessment:** Sentinel-1 radar data in VH polarization (IW mode) are processed to analyze flood extent

and permanent water in a defined ROI. Pre- and post-flood images are compared and are subjected to Lee filtering of the noise. The flooded areas are highlighted in red, permanent water is highlighted in blue, and the area flooded is calculated using a -20 threshold. Flood hazard assessment considered distance from water, topography (TPI), vegetation (NDVI), moisture (NDWI), and elevation. Flooded areas vary in risk accordingly a hazard map from these factors is created which classifies the risk over a gradient from low to high.

- **Disaster Prediction:** Long Short-Term Memory (LSTM) model uses historical weather and hydrological data in the dataset to adequately predict floods. The key features are temperature, precipitation, sunshine duration, wind speed, and river discharge. The model is trained to predict the occasion of flooding. Predictions are made allowing the model to predict the likely occurrence of floods by considering historical data and current data of the period of 7 days.

3.4 Analytics and Reporting

The report that SAFER provides depends on the area requested by the user. The data includes the cause of the disaster, damage occurred, casualties, safe regions and danger zone. Using these parameters, the seriousness of the disaster will be calculated. The report will also include a map highlighting the affected regions. If the user wishes, they can print the report, which is generated using the html2pdf.

3.5 Voice-Controlled Geospatial Map

The voice-controlled interface for geospatial map allows hands-free navigation and data retrieval by use of voice commands. The system employed speech recognition and natural language processing to generate intention from the given command patterns like “show flood zones”.

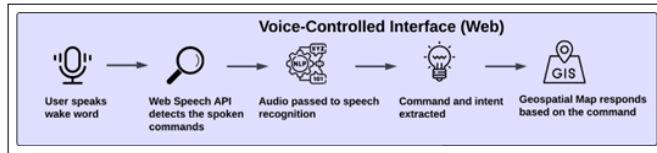


Fig. 2. Voice-Controlled Interface for Geospatial Map

The Fig. 2 represents the workflow of a voice-controlled interface. This begins with the user speaking a wake word. The Web Speech API takes the correct commands spoken out loud which is then handed over to the NLP engine that extracts its intent. Based on the extracted intent, the geospatial map presents the appropriate response.

3.6 Intelligent Ambulance Booking System

SAFER provides a quick booking of an ambulance with a single click. The system automatically looks up the user’s location, passing it on to the nearest hospitals and rescue teams in the direction of the injured victim. The ambulance’s and the

user’s positions are traced on the map using the Google Maps API. If the victim cannot access internet, SAFER will send the ambulance request through an SMS using the Twilio service.

IV. RESULT AND ANALYSIS

Through the implementation of SAFER the disaster management of Northeast India has made several strides forward towards enhanced situational awareness and preparation for disasters. The results also indicate that this system significantly decreases emergency response times and increases resilience in general scenarios.

4.1 Dataset

The dataset used for flood assessment include several satellite and vector datasets were utilized. The Global Surface Water Dataset, SRTM Digital Elevation Model, Sentinel-2 Surface Reflectance and Copernicus Sentinel-1 GRD with SAR imagery to detect flood areas and surface reflectivity changes. For prediction open-meteo API offers data on river discharge for flood risk assessment. The merged data file contains 2–3 days of observations from 16 locations across India covering historical disaster periods, aligned by location and date. This dataset includes 40 records and 18 attributes that provide information about climatic and hydrological interactions during disasters.

4.2 Result and Analysis

In this section, we will present and analyze the outcomes related to the elaboration of assessment maps and tools for flood assessment and risk evaluation. Attention has been directed toward analyzing the river discharge patterns; then the flood-affected areas, risk areas, and integration of voice-controlled mapping systems will be studied in depth.

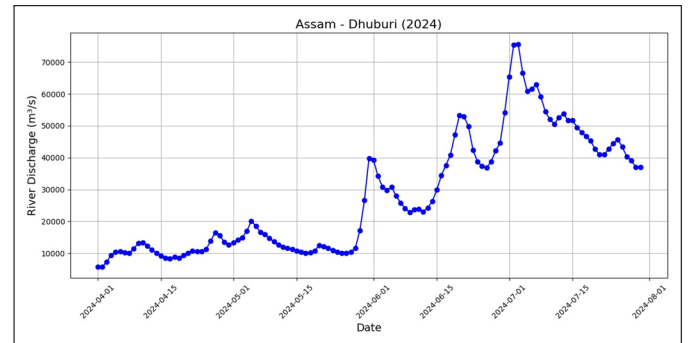


Fig. 3. Assam – Dhuburi (2024) River Discharge

The river discharge trends for Dhuburi, Assam (2024) depicted in Fig. 3 are derived from the data of Open Meteo API. Until mid-May, the rates stay below 20,000 m³/s and reach 40,000 m³/s at the end of May. In early July it peaked at over 70,000 m³/s due to heavy rains and upstream contributions. This is necessary data for disaster management. It enables early flood warnings, better planning of evacuations, and distribution of essential resources to minimize losses. Analyzing yearly trends will help in

identifying the periods of high rainfall and frequent floods, which will improve disaster preparedness and mitigation.

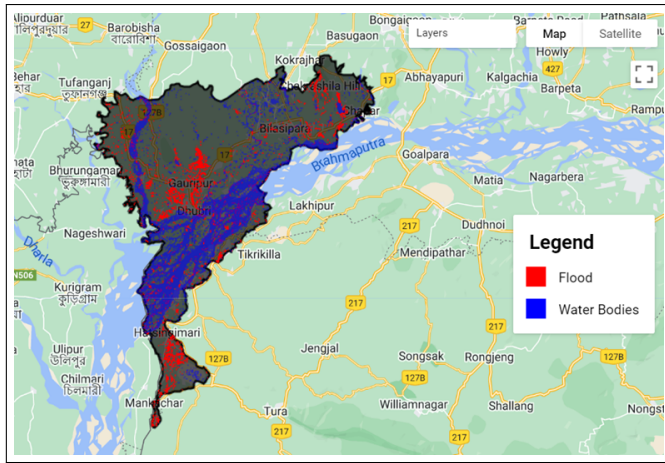


Fig. 4. Flood Assessment Map (Google Earth Engine)

Fig. 4 identifies these risk zones in Dhuburi for July 2024. The affected areas were identified by analyzing Sentinel-1 images through pre- and post-flood image differencing. This action helps with evacuation planning, resource allocation, and flood defenses improvement.

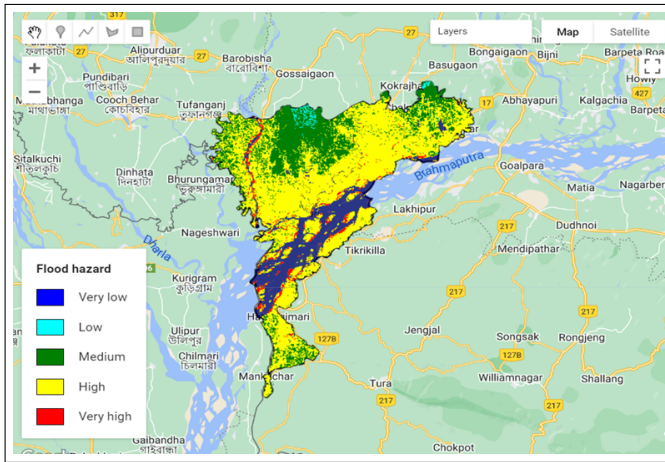


Fig. 5. Flood Risk Map (Google Earth Engine)

Fig. 5 shows the flood risk map created using Google Earth Engine focuses on Dhuburi. It identifies permanent water bodies using the Global Surface Water (GSW) dataset and assessing its proximity through a distance transform. Elevation data from SRTM DEM and Topographic Position Index (TPI) highlight areas prone to floods. Vegetation and wetness were analyzed using Sentinel-2 satellite imagery, and each factor (distance, elevation, topography, vegetation, and wetness) was scored from 1 to 5 to create a flood hazard index, color-coded from very low to very high risk.

The Fig.4 and Fig.5 identifies flood-prone areas for the assessment of disasters and allocation of resources, permitting authorities to put the attention on areas with

greater risks and work on planning evacuation routes as well as shelters. The people are informed about the safety of an area, allowing them to prepare for floods by stocking up on basic necessities, protecting their possessions, and planning evacuations ahead of time. It provides long-term resilience through guiding decisions on infrastructure development and flood mitigation strategies.

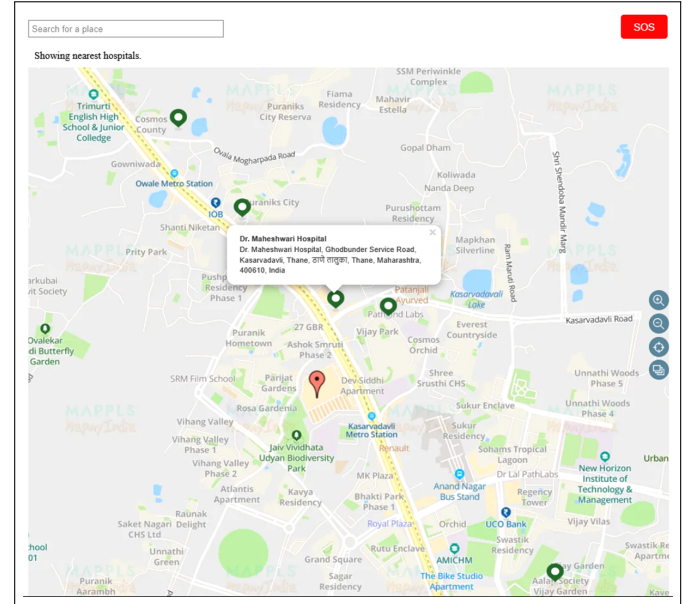


Fig. 6. Voice-Controlled Map Website: Locate Nearest Hospital

Fig. 6 displays a geospatial map with voice-controlled functionality. For which a TTS system has been developed for voice commands to the application and a STT system to interpret those user commands. The voice-controlled map system enables users' hands free navigation. This is extremely beneficial during floods and landslides as this can allow someone to get essential information as quickly as possible. During such emergencies, all a user has to do is issue voice commands such as "show nearest hospitals" and the routes are provided to them.

V. CONCLUSION

The SAFER project enhances disaster management in Northeast India by using Sentinel-1 and Sentinel-2 satellite data for flood assessments and LSTM-based machine learning models for disaster prediction. It provides voice-controlled geospatial maps for emergency services, an intelligent ambulance booking system with GPS tracking, and a voice-guided first aid assistant. The project generates reports to improve disaster strategies and resource planning. Through these features, it improves response times, aids in resource allocation, and enhances the efficiency of emergency services. Future plans include disaster evacuation systems, integration of live satellite imagery and IoT sensors, multi-linguistic real-time alerts, and collaboration with emergency services for timely and effective responses.

REFERENCES

- [1] P. Anbarasu, S. Arunkumar, R. Gowtham, S. Deepak, and M. Latha. The ambulance booking and tracking system. *International Research Journal of Modernization in Engineering Technology and Science*, 6(4):10287, 2024.
- [2] Vivek Chamoli, Rishabh Bahuguna, R Gowri, Rishi Prakash, Anurag Vidyarthi, and Ved Prakash Dubey. Landslide detection in uttarakhand region using active remote sensing. In *2024 2nd International Conference on Device Intelligence, Computing and Communication Technologies (DICCT)*, pages 154–157. IEEE, 2024.
- [3] Enrico Collini, LA Ipsaro Palesi, Paolo Nesi, Gianni Pantaleo, Nicola Nocentini, and Ascanio Rosi. Predicting and understanding landslide events with explainable ai. *IEEE Access*, 10:31175–31189, 2022.
- [4] Dhyan Dhaval Desai, Joyeeta Dey, Sandeep Kumar Satapathy, Shruti Mishra, Sachi Nandan Mohanty, Pallavi Mishra, and Sandeep Kumar Panda. Optimal ambulance positioning for road accidents with deep embedded clustering. *IEEE Access*, 11:59917–59934, 2023.
- [5] Jinyang Du, John S Kimball, Justin Sheffield, Ming Pan, Colby K Fisher, Hylke E Beck, and Eric F Wood. Satellite flood inundation assessment and forecast using smap and landsat. *IEEE journal of selected topics in applied earth observations and remote sensing*, 14:6707–6715, 2021.
- [6] Kuldeep, Pradeep Kumar Garg, Rahul D Garg. Geospatial techniques for flood inundation mapping. In *2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 4387–4390. IEEE, 2016.
- [7] Omid Ghorbanzadeh, Hejar Shahabi, Alessandro Crivellari, Saeid Homayouni, Thomas Blaschke, and Pedram Ghamisi. Landslide detection using deep learning and object based image analysis. *Landslides*, 19(4):929–939, 2022.
- [8] R. Jyothsingh, Thota Swathi, Kareti Lepakshi, Pulivarthi Akash, and Puvvaladasu Hemanth. Ambulance booking system. *International Journal of Innovative Science and Research Technology (IJSRT)*, 9(4):2033–2039, 2024.
- [9] R Keerthana, T Ananth Kumar, P Manjubala, and M Pavithra. An interactive voice assistant system for guiding the tourists in historical places. In *2020 International Conference on System, Computation, Automation and Networking (ICSCAN)*, pages 1–5. IEEE, 2020.
- [10] Homeyra Mahmoudi, Silvana Camboim, and Maria Antonia Brovelli. Development of a voice virtual assistant for the geospatial data visualization application on the web.
- [11] S Praharaj and E Wentz. Building community resilience through geospatial information dashboards. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, 48:151–157, 2022.
- [12] Supattra Puttinaovarat and Paramate Horkaew. Flood forecasting system based on integrated big and crowdsourced data by using machine learning techniques. *IEEE Access*, 8:5885–5905, 2020.
- [13] Seema Redekar, Chirag Gupta, Sneha Baital, Saloni Kharat, and Tarun Iyer. Ambulo mobile application for online ambulance booking. In *2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC)*, pages 174–181. IEEE, 2024.